

Heuristic Algorithms

In computer science, a heuristic is a problem-solving method that uses incomplete information to derive a potentially inaccurate or imprecise solution.

HOW DOES HEURISTICS WORK?

In optimized search, heuristic algorithms refine each iterative result until they find the result closest to the solution. In other words, typical heuristic search algorithm uses a search tree, qualitatively selecting branches more likely to produce the result closest to the solution by assessing the given information.

1 At every decision point in the search tree, the algorithm applies the sum of all that it has learnt from previous decision points, and by eliminating the unlikely, refines the solution to the extent possible with the given information, or most likely result.

In this scenario, the algorithm is trying to find the most likely candidate for Mario from the crowd, who resembles him the closest.

2 There are hundreds of types of heuristic optimized search, with many variations. It can even move both upstream and downstream to evaluate the candidate solution.

The algorithm, in our example, already has established several points of contact -- or likely Mario candidates, that it traverses up and down its logic tree.

3 Another major application of heuristics are evolutionary algorithms. An evolutionary algorithm is a type of a heuristic optimization algorithm which mimics organic evolution, by using reproduction, mutation, recombination, and selection as devices to find the optimal configuration for a specific system.

4 So the way heuristics works finally, is it searches the crowd image for potential Mario-like individuals. It tries to match them by hue, size, distinct characteristics like moustache, cap, etc. Once it has arrived at a likely match, it spots the individual and highlights it. The algorithm also learns from its discovery of the result and saves the methodology.

